Gummy Bear Transfer — OKR/SMART Phase-Gate Roadmap

Project: SRT embodied-AI · Robot: LeRobot SO-101 (leader + follower) · Task class: deformable-object manipulation · Framed 2026-09-03 · Method: OKR + SMART + phase-gates

Objective (the claim / direction)

Demonstrate data-efficient manipulation of a small, deformable object (a gummy bear) on the SO-101 arm — i.e. reach a reliable transfer task with the **fewest** teleop episodes, comparing ACT vs Diffusion Policy vs SmolVLA.

Key Results (measurable)

KR1	≥90% transfer success over 50 trials (bear fully in container B, no drop) by the Dec SRT review — % success over N=50
KR2	Reach KR1 in ≤150 teleop episodes (the "less data" claim) — episodes-to-success
KR3	Rigid-block baseline: ≥80% over 40 trials, 2 weeks in (de-risk) — % success over N=40
KR4	Publish the episodes⟶success curve for ACT vs Diffusion vs SmolVLA — plotted curve + table

North-star metric: episodes to reach the target success rate (data-efficiency at fixed success). Secondaries: success rate, robustness across unseen configs, safety (no hardware damage).

Phase-Gate Roadmap

Phase	Goal and gate-metric	Status	Blocker
Phase 0, Env + hardware gate, (Basic Exp 2)	Ubuntu host + SO-101 arms + camera. **Gate:** clean teleop + stable camera; 1 rigid episode recorded.	⚠️ Needs hardware (software env done)	Physical arm + Ubuntu (lead-time)
Phase 1, Rigid-block baseline, (Exp 1)	Grasp a hard cube A⟶B, collect ≥50 episodes. **Gate:** ≥80% / 40 trials; dataset pushed.	❌ Not started	Phase 0
Phase 2, Gummy-bear data protocol	Define task (A⟶B), camera, lighting, randomization; budget ≤150. **Gate:** reproducible recording, ≥50 episodes.	❌ Not started	Phase 1
Phase 3, Policy training, (Exp 3)	ACT then Diffusion Policy (from-scratch BC). **Gate:** eval improves on held-out configs, no crashes.	❌ Not started	Phase 2
Phase 4, Data-efficiency comparison, (Exp 4 + 6 — the research)	SmolVLA vs ACT vs Diffusion: episodes-to-success. **Gate:** a method hits KR1 in ≤150 episodes; curve produced.	❌ Not started	Phase 3
Phase 5, Hardening + real "complex"	Randomize start/container/lighting; the deformable+translucent difficulty is the claim. **Gate:** robust across unseen configs; quantify failure modes.	❌ Not started	Phase 4

Note: Phases 0–4 are the de-risked path; Phase 5 is where "complex" gets **earned**, not assumed. The gummy bear's hardness (translucent + deformable + sticky) is the research value — prove the pipeline on a rigid block first.

Risk register

Risk	Likelihood	Mitigation
Phase 0 lead-time (arm / Ubuntu / USB / power)	High	Order + set up early; it gates everything
Translucent-object perception fails	High	Start rigid (Phase 1); backlight/depth; or a colored bear
Deformable grasp (no rigid pose, sticky)	Med	Soft/geometric grasp; scoop; record teleop "how to grab it"
Data too expensive (slow on real arm)	Med	Domain randomization, augmentation, pre-trained VLA (SmolVLA)

Sim-to-real / calibration drift	Med	Calibrate per session; keep camera+lighting fixed during collection
Scope creep to handshake/humanoid (Exp 5/7)	High	Anchor to the gummy-bear objective; park Exp 5/7

Data-budget table (the thesis number)

Method	Episodes to ≥90% (target)	Episodes collected	Success @ eval
ACT (from-scratch)	TBD	TBD	TBD
Diffusion Policy	TBD	TBD	TBD
SmolVLA (pretrained)	TBD	TBD	TBD

Fill in as you train. The "less data" result = the SmolVLA row beating ACT/Diffusion's episode count at the same success.

Milestones (SMART, time-bound)

- **M0 (Phase 0):** By 3 weeks, Ubuntu + SO-101 calibrated, teleop + a recorded episode working.
- **M1 (Phase 1):** By 2 weeks after M0, ≥80% rigid-block success over 40 trials + dataset pushed.
- **M2 (Phase 2+3):** By 4 weeks after M1, ≥50 gummy-bear episodes + a trained ACT policy that improves on eval.
- **M3 (Phase 4):** By the Dec SRT review, KR1 + KR2 met; episodes⟶success curve produced (the ICRA-style result).

Already done vs. needed (quick view)

Done: software env (`conda lerobot`, CUDA torch, LeRobot 0.6.1 + CLI tools, gymnasium, jupyter); pipeline understood + Code 1 verified; study material staged; proposal + SO-101 + SmolVLA tutorials read.

Needed (in order): physical SO-101 + Ubuntu (Phase 0) ⟶ rigid-block baseline (Phase 1) ⟶ gummy-bear data (Phase 2) ⟶ ACT/Diffusion (Phase 3) ⟶ SmolVLA data-efficiency (Phase 4) ⟶ hardening (Phase 5).

This roadmap is the artifact; keep it updated at each gate.